Circular Reasoning and Unsupported Statements
Answer Key

High School Geometry

Quick Answers

1. —	3. 25, 85	5. 23, 103
2. —	4. 4, 11	6. —

Curriculum

Level: High School Geometry
HSG-CO.A–HSG-CO.D – *problems 1, 2, 3, 4, 5, 6*
Difficulty 2–5 of 5 across 9 tagged checks.

Common wrong answers

3. *If they got 70: set the two angles equal on an unsupported corresponding-angles claim, getting x = 20 and an angle of 70 degrees*
4. *If they got 13: assumed M was a midpoint, which no given supports, solving 2x + 3 = 4x - 7 for x = 5 and reporting AM = 13*
5. *If they got 38: called the two angles vertical angles and set them equal, getting x = 10 and an angle of 38 degrees*

1. The circular step. It is the middle step: the statement to be proved is "vertical angles are congruent", and its reason is "vertical angles are congruent". The proof therefore assumes exactly what it was asked to establish, so it establishes nothing. [Manual review — look for the student naming the middle step and the reuse of the conclusion as its own reason.]
A correct proof uses linear pairs instead. Since A, Q, B and C, Q, D are straight lines, $\angle AQC$ and $\angle AQD$ form a linear pair, and so do $\angle BQD$ and $\angle AQD$. Both $\angle AQC$ and $\angle BQD$ are therefore supplementary to the SAME angle, $\angle AQD$. Angles supplementary to the same angle are congruent (congruent supplements), which gives $\angle AQC \cong \angle BQD$ — with no step that cites the conclusion.

2. Which reason works? Reason (i), the Segment Addition Postulate, is the one that supports the statement: it says precisely that when M is between A and C, the two short lengths add to the whole.
Reason (ii) is circular — it offers the statement itself as its own justification, so it carries no information.
Reason (iii) is unsupported: nothing in the givens says M is a midpoint. Midpoint is a stronger claim (it would force $AM = MC$), and a proof may not help itself to a property it was not given. Note it is not merely unnecessary; it is unavailable. [Manual review — credit an answer choosing (i) and distinguishing "repeats itself" from "assumes an extra given".]

3. Dana's mistake. (a) "Corresponding angles are congruent" is a statement about two parallel lines cut by a transversal. The givens describe one line with a ray on it — a linear pair. No parallel lines are given, so the reason is unsupported (and the two angles here are not corresponding angles at all). [Manual review — the explanation must name the missing parallel-lines given.]
(b) A linear pair is supplementary, so add rather than equate:

$$(3x + 10) + (5x - 30) = 180$$

$8x - 20 = 180$, so $8x = 200$ and $\boxed{x = 25}$.
Then $m\angle AQB = 3(25) + 10 = 85$, so $\boxed{m\angle AQB = 85^\circ}$. Check: $m\angle BQC = 5(25) - 30 = 95$, and $85 + 95 = 180$ as a linear pair requires. Dana's 70 fails that check — her $x = 20$ gives 70 and 70, which sum to 140.

4. Ben's mistake. (a) He would need "M is the midpoint of $\overline{AB}$" as a given (or a proved statement). The givens supply only that M is between A and B together with the three lengths, so $AM = MB$ is an extra assumption, not a consequence. [Manual review — the answer must say the midpoint given is absent.]
(b) Betweenness gives the Segment Addition Postulate:

$$(2x + 3) + (4x - 7) = 20$$

$6x - 4 = 20$, so $6x = 24$ and $\boxed{x = 4}$.
Then $AM = 2(4) + 3 = \boxed{11}$, and $MB = 4(4) - 7 = 9$. Check: $11 + 9 = 20$, which matches the given AB. Ben's answer of 13 for AM would force $MB = 13$ and $AB = 26$, contradicting a given — the fastest way to catch an unsupported assumption is to test it against the givens you do have.

5. Priya's mistake. (a) Vertical angles are the opposite pairs formed by two intersecting lines, and they share only their vertex. $\angle AQC$ and $\angle AQD$ share the whole ray QA, so they are adjacent, not vertical. Because C, Q, D are collinear, their non-shared sides form a straight line: they are a linear pair, hence supplementary. [Manual review — credit "they share a side" plus the correct linear-pair naming.]
(b) Supplementary means the measures add to 180:

$$(5x - 12) + (3x + 8) = 180$$

$8x - 4 = 180$, so $8x = 184$ and $\boxed{x = 23}$.
Then $m\angle AQC = 5(23) - 12 = 103$, so $\boxed{m\angle AQC = 103^\circ}$. Check: $m\angle AQD = 3(23) + 8 = 77$, and $103 + 77 = 180$. Priya's $x = 10$ gives 38 and 38, a total of 76 — nowhere near a straight angle, which is the signal that the assumed relationship was the wrong one.

6. Repairing the isosceles argument. (a) The argument proves $\overline{CA} \cong \overline{CB}$ from "the triangle is isosceles", and then proves "the triangle is isosceles" from the congruent base angles by citing the Isosceles Triangle Theorem — whose own statement is *if two sides are congruent, then the base angles are congruent*. So the second half assumes the very side congruence the first half was supposed to establish. The reasoning runs in a loop and never touches the given. [Manual review — the student must identify the loop, not merely call the argument wrong.]
(b) A correct argument uses the CONVERSE of the base-angles theorem, which is a separately proved theorem:

Statements	**Reasons**
$\angle A \cong \angle B$ in $\triangle ABC$	Given
$\overline{CA}$ is opposite $\angle B$; $\overline{CB}$ is opposite $\angle A$	Definition of the side opposite an angle
$\overline{CA} \cong \overline{CB}$	Converse of the Isosceles Triangle Theorem (sides opposite congruent angles are congruent)
$\triangle ABC$ is isosceles	Definition of an isosceles triangle

Note the order: "isosceles" is now the CONCLUSION, drawn from the side congruence, instead of a premise smuggled in to prove it.